Term Project - Checkpoint A

Machine Learning Enhanced Adaptive Momentum ETF

A Literature Review and Proposed Quantitative Fund Strategy

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1. Introduction - Why This Research?

The research question for this project is: Can a simple, diversified momentum strategy be improved by using machine learning to reduce equity exposure before periods of elevated market risk? I propose a hypothetical, actively managed exchange-traded fund (ETF), the Machine Learning Enhanced Adaptive Momentum ETF (the "Fund"), to study this question. The investment objective is to pursue long-term, risk-adjusted growth with explicit trading and risk-control rules. This is a proposed research strategy, not an existing fund or an offer to invest.

The project is motivated by two practical problems. First, passive equity exposure can experience substantial drawdowns. Second, a more complex prediction model is not automatically better than a transparent trading rule. The Fund therefore uses a clear momentum and trend-following strategy as a baseline and adds a machine-learning (ML) downside-risk model as a required research component. The goal is to test whether the ML overlay makes a measurable difference, not to assume it improves performance.

The intended users are investors seeking a systematic, explainable alternative to static buy-and-hold allocations, as well as analysts and portfolio managers interested in auditable investment rules. The project will produce a research report, reproducible data workflow, explicit portfolio rules, and a backtest comparing three approaches: SPY buy-and-hold, a rules-based momentum portfolio, and the same portfolio with ML-based risk management. Investors must recognize that any of these approaches could underperform a passive benchmark and that a defensive allocation cannot eliminate losses.

Investment philosophy and prospectus summary

The Fund follows a long-only, rules-based approach. It seeks assets with positive medium-term momentum and supportive long-term trends, rebalances once per month, and directs unused capital to short-term U.S. Treasury exposure. It does not rely on discretionary stock selection, company valuation estimates, leverage, short selling, or options hedges. The Fund is not buy-and-hold: holdings may change monthly, though it is designed to trade less frequently than a daily strategy.

The Fund also includes an ML-enhanced version as a central part of this research. A supervised-learning model will estimate the probability of a sharp SPY decline over the next calendar month. When that probability exceeds a specified threshold, the strategy will reduce equity ETF exposure and move the difference into short-duration Treasury exposure. The ML version will be implemented and evaluated even if its predictions are weak; any improvement remains a research hypothesis.

2. Literature Review - Who Has Studied Related Approaches?

Diversification and portfolio construction

Markowitz (1952) established the importance of evaluating investments as a portfolio rather than selecting assets only by individual expected return. His framework emphasizes how correlations affect overall portfolio risk. This supports using a mix of broad equities, sector equity ETFs, Treasuries, and a gold-linked product. However, the initial Fund does not claim to solve a mean variance optimization problem: it uses a deliberately simple ranking-and-cap approach that can later be compared with optimized weights.

Momentum, trend following, and tactical allocation

Moskowitz, Ooi, and Pedersen (2012) documented time-series momentum across liquid futures markets, including equities, bonds, currencies, and commodities. Their results motivate studying whether past returns contain useful trend information; they do not demonstrate that the proposed ETF selection rule will outperform. Faber (2013, updated working paper) examined systematic tactical asset allocation using trend rules across asset classes. Together, these studies support an interpretable baseline based on prior returns and moving averages, but they leave open important questions about implementation timing, sample choice, and transaction costs. Fundamental analysis emphasizes business value and financial statements, while technical analysis uses price trends. I chose technical signals because the Fund needs consistent monthly rules for equity, Treasury, and gold instruments.

Risk management and the limits of prediction

Moreira and Muir (2017) studied portfolios that scale down risk during high-volatility conditions, providing motivation for adjusting exposure instead of trying to predict daily price direction precisely. Gu, Kelly, and Xiu (2020) found that machine-learning methods can capture nonlinear relationships in asset-pricing data, with signals that include momentum and volatility. Their work supports testing nonlinear models, not assuming that they will work for a small universe of monthly ETF observations.

There is a tension between the approaches. Simple trend rules are easier to audit and less demanding of data; machine learning may detect interactions, but can also overfit noisy, nonstationary returns. A credible comparison must hold the investment universe, execution timing, evaluation periods, and transaction-cost assumptions constant. The Fund therefore treats transparency and out-of-sample evaluation as central design requirements.

Active ETF structure

An actively managed ETF may select and trade holdings based on a stated investment objective rather than tracking an index (U.S. SEC, 2023). This makes an ETF structure conceptually suitable for the proposed monthly investment rules. Most ETF shares trade during exchange hours, rather than automatically trading 24/7. The proposed Fund is unlevered and long-only. One proposed holding, GLD, is a gold-backed exchange-traded product rather than a conventional registered

investment-company ETF; this distinction will be made clear in the investment-universe documentation.

3. Methods - How Will the Fund Work?

3.1 Investment universe and portfolio structure

The Fund will analyze six listed instruments, but it is not required to hold all six at the same time. SPY, XLF, XLV, IEF, and GLD form the five-asset ranking universe; SHY is the defensive holding. In any given month, the portfolio will select at most two qualifying instruments, with the unused weight assigned to SHY. All six remain part of the data and investment-universe research. The backtest will first check inception dates, price availability, and the suitability of corporate action adjusted prices.

Ticker	Exposure	Purpose
SPY	U.S. large-cap equities	Core equity market
XLF / XLV	U.S. financial / health-care equities	Diversification across equity sectors
IEF	Intermediate-term U.S. Treasuries	Bond exposure
GLD	Gold-linked exchange-traded product	Alternative asset exposure
SHY	Short-term U.S. Treasuries	Defensive, cash-like investment; not literal cash

Because GLD began trading in 2004 and the funds have different inception dates, the common universe backtest is expected to start no earlier than the mid-2000s, after enough history is available for lagged indicators. The actual date will be fixed after inspecting adjusted daily price histories. No unavailable early returns will be filled with invented data.

3.2 Rule-Based Momentum Trading Strategy

Rule	Proposed implementation
Decision time	At the final trading-day close of each calendar month, compute signals using information available through that close.
Eligibility	For SPY, XLF, XLV, IEF, and GLD: total-return proxy price > prior 200-trading-day simple moving average AND trailing 126-trading-day return > 0.
Ranking	Rank eligible ETFs by their trailing 126-trading-day total return, highest to lowest; break ties alphabetically by ticker.
Allocation	Select at most two eligible ETFs. Assign 50% weight to each selected ETF; assign each unused 50% slot to SHY.
Execution	Submit rebalancing orders at the next trading session open or a consistently modeled executable price; never assume a trade at the already-observed signal close.

Rule	Proposed implementation
Constraints	Fully funded, long-only; no borrowing, leverage, short selling, derivatives, or discretionary overrides. No individual selection exceeds 50%.
Costs	Apply an assumed 10 basis points per dollar of turnover and compare sensitivity at 0 and 20 basis points. ETF internal costs remain reflected in observed prices.

The rule may hold both selected ETFs, just one ETF plus SHY, or 100% SHY when no candidate qualifies. Sector ETFs can still be correlated with SPY; diversification will be assessed rather than assumed. Risk assets here refer primarily to equities and gold, while Treasury ETFs also retain interest-rate risk.

3.3 Machine Learning-Based Risk Management Strategy

The ML-enhanced portfolio is a required second strategy for the final project, not an optional extension. Its supervised-learning target will be whether the next full calendar-month SPY return is below -3%. Explanatory variables available at each month-end will include trailing SPY returns over 1, 3, and 6 months; trailing 21-day annualized SPY realized volatility; SPY distance from its 200-day moving average; and recent returns for IEF and GLD relative to SPY. The final feature definitions and missing-data treatment will be determined from the development period without looking at the final test results.

I will implement a regularized logistic regression as an interpretable ML benchmark and XGBoost as a nonlinear ML model. I will compare their risk probabilities using walk-forward or expanding-window validation, selecting the model and its hyperparameters on development data only. Each future-month target must be completely observed before it can be used for training; chronological gaps will be applied where necessary to prevent overlapping-label leakage. The most recent period will be reserved as an untouched hold-out test. A provisional 0.60 probability threshold will be specified before testing; any later adjustment must be performed only within development folds.

The proposed ML-enhanced allocation rule is explicit: if the selected classifier estimates a next-month downside probability greater than 0.60, halve the baseline weights in equity ETFs (SPY, XLF, and XLV) and transfer the removed weight to SHY. IEF and GLD weights remain unchanged. Otherwise, retain the baseline portfolio weights. The ML portfolio will be coded and reported alongside the rules-only version, even if it has worse accuracy or risk-adjusted returns. A model that does not outperform the baseline will be reported as a negative research finding rather than removed from the project.

3.4 Hedging, invested capital, and expected turnover

The Fund uses defensive rotation rather than a direct options or short-sale hedge. In a downturn it may increase SHY holdings, but SHY is a short-duration Treasury ETF and not a guaranteed capital-protection instrument. The allocation rule invests nearly all model capital in ETFs at each monthly rebalance; literal cash is expected to remain near zero except for operational residuals. The portion invested in growth-oriented ETFs versus defensive SHY can vary from 0% to 100% and

cannot responsibly be assigned a 25-year historical average before testing. I will measure average SHY weight, growth-asset weight, time spent 100% defensive, and annual portfolio turnover over the available common-history period. If a full 25-year horizon is required, I will identify suitable historical index proxies and disclose any differences from the ETF backtest.

The portfolio is actively managed with a scheduled monthly review (up to 12 regular rebalance dates per year). Actual annual turnover will depend on how often rankings and eligibility change. The backtest will report annual turnover as the sum of absolute changes in portfolio weights, and evaluate whether trading costs outweigh benefits.

3.5 Testing plan for later checkpoints

At Checkpoint B, I plan to retrieve and clean adjusted daily price histories, check ticker coverage and corporate-action adjustments, implement the baseline allocation engine, and freeze signal-to-execution timing. The project will implement the ML classifier and compare (1) SPY buy-and-hold, (2) the rules-based portfolio, and (3) the ML-enhanced portfolio. Evaluation will include compound annual growth rate, annualized volatility, Sharpe ratio, maximum drawdown, cumulative performance, turnover, average SHY allocation, and net performance after transaction costs. The ML model will also be evaluated for classification quality, probability calibration, and stability across time. These are planned analyses; Checkpoint A does not report actual backtest or model-performance results.

4. Results - What Has Been Learned So Far?

Checkpoint A findings are conceptual results from the literature review and strategy design, not investment returns or backtest outputs. Four insights changed the initial idea:

- First, the Fund needs a transparent reference strategy. Momentum and a long-term trend filter provide explicit buy-and-sell rules that can be implemented before any complex model is introduced (Moskowitz et al., 2012; Faber, 2013).
- Second, portfolio construction matters as much as the prediction model. Correlation across SPY and equity sectors can reduce the diversification expected from holding several tickers, so the later evaluation must examine co-movement and drawdowns (Markowitz, 1952).
- Third, an ML model should be used for a specific decision: whether to cut equity exposure when downside risk appears elevated. This is more testable than a vague claim that AI will predict stock prices (Moreira & Muir, 2017; Gu et al., 2020).
- Fourth, apparent performance can be misleading without an untouched test set, delayed execution, and transaction costs. A simple strategy that survives these checks may be preferable to a more complex model with stronger in-sample results.

Based on this review, I narrowed the original idea to a monthly, six-instrument, unlevered strategy and made ML-based downside-risk management a required comparison against an interpretable rules-based baseline. The ML component has a specific task and a specific portfolio response, so it can be evaluated even if it fails. I have not yet estimated investment returns, drawdowns, turnover, model accuracy, or a 25-year average allocation. Those numbers will be produced only after the data and backtesting program have been checked.

5. Conclusions - So What, and What Comes Next?

The literature supports a feasible research design for an actively managed quantitative investment strategy that combines momentum rules with an ML risk overlay. It does not establish that the proposed portfolio will earn excess returns or that ML will improve downside protection. The main contribution at Checkpoint A is a transparent trading specification, an ML-enhanced version, and a test plan that can distinguish a useful risk signal from a complicated but ineffective model.

The largest concerns are (1) the small number of monthly observations available to train an ML classifier, (2) nonstationarity and changing market regimes, (3) look-ahead bias from overlapping labels or trading at already-known prices, (4) costs and ETF differences in trading history, and (5) possible concentration in correlated equity funds. No rule guarantees protection in a sudden crash, and ML predictions may be uninformative.

At Checkpoint B, I will assemble reliable price data and implement the baseline trading engine while preparing the monthly ML dataset and validation design. I will then implement the logistic and XGBoost classifiers, apply the risk overlay, and evaluate all three portfolio alternatives on common dates and costs. By Week 10, the report will show both successes and failures and explain whether ML added value over transparent momentum rules.

References

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